

# Quantum Fault Tolerance Meets Toom's Contours

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## 1 Hadamard and S Gates

**Definition 1.1** (The  $g_a \leftrightarrow g_b$  Transport). Let  $d, d'$  be integers,  $\mathcal{G}$  be a  $(d, d')$ -Toric complex, generated by generator set  $S$ . Since the vertices in  $\mathcal{G}$  are the elements of the abelian group generated by  $S$ , each of vertex  $v \in \mathcal{G}$  has a unique presentation of:

$$v = \prod_{i \in [d]} (g_i)^{x_i}$$

For generators  $g_a, g_b$  in  $S$ , we define the  $g_a \leftrightarrow g_b$  transport map:  $\phi_{g_a \leftrightarrow g_b} : \mathcal{G} \rightarrow \mathcal{G}$  as:

$$\phi(v)_{g_a \leftrightarrow g_b} := \prod_{i \in [d] \setminus \{a, b\}} (g_i)^{x_i} g_a^{x_b} g_b^{x_a}$$

$$\phi_{g_a \leftrightarrow g_b}(vu) = \phi_{g_a \leftrightarrow g_b}\left(\prod_{i \in [d]} (g_i)^{x_i + y_i}\right) = \prod_{i \in [d] \setminus \{a, b\}} (g_i)^{x_i + y_i} g_a^{x_b + y_b} g_b^{x_a + y_a} = \phi_{g_a \leftrightarrow g_b}(v) \phi_{g_a \leftrightarrow g_b}(u)$$

$$\begin{aligned} \phi(\theta_{v, S'})_{g_a \leftrightarrow g_b} &= \phi\left(\bigcup_{I \subset S'} \left\{ \left( \prod_{g \in I} g \right) v \right\}\right)_{g_a \leftrightarrow g_b} \\ &= \left( \bigcup_{I \subset S'} \left\{ \phi\left( \left( \prod_{g \in I} g \right) v \right)_{g_a \leftrightarrow g_b} \right\} \right) \\ &= \left( \bigcup_{I \subset S'} \left\{ \phi\left( \left( \prod_{g \in I} g \right) v \right)_{g_a \leftrightarrow g_b} \right\} \right) \\ &= \theta_{\phi(v)_{g_a \leftrightarrow g_b}, \phi(S')_{g_a \leftrightarrow g_b}} \end{aligned}$$

So the map sends bits to bits and checks to checks.

**Claim 1.2.** For  $d' = d/2$ , the  $H$  gate, can be computed transversally.

*Proof.* The action of the permutation  $g_a \leftrightarrow g_b$ ,  $v \mapsto v$ , sends checks to checks.

$$\theta_{\mathbf{v}, \mathbf{S}'} \rightarrow \theta_{\mathbf{v}, \phi(\mathbf{S}')}$$

And:

$$\begin{aligned} \partial^+ \phi(\theta_{\mathbf{v}, \mathbf{S}'}) &= \sum_{g \in S/\phi(S')} \theta_{\phi(\mathbf{v}), \phi(\mathbf{S}') \cup \mathbf{g}} + \sum_{g \in S/\phi(S')} \theta_{\phi(\mathbf{g}^{-1}\mathbf{v}), \phi(\mathbf{S}') \cup \mathbf{g}} \\ &= 0 \\ &= \partial \left( \sum_{g \in S/\phi(S')} \theta_{\mathbf{v}, \phi(\mathbf{S}') \cup \mathbf{g}} + \sum_{g \in S/\phi(S')} \theta_{\mathbf{g}^{-1}\mathbf{v}, \phi(\mathbf{S}') \cup \mathbf{g}} \right) \\ &= \sum_{g \in S/S'} \theta_{\mathbf{v}, \mathbf{S}' \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{v}, \mathbf{S}' \cup \mathbf{g}} = 0 \end{aligned}$$

□

**Claim 1.3.** The  $S$  gate can be computed transversally.

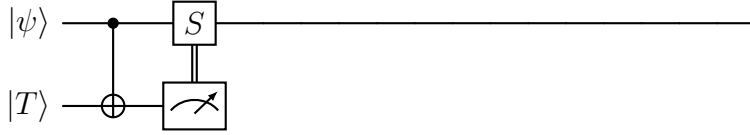


Figure 1: Simulating the  $T$ -gate, using the  $|T\rangle$  magic state.

## 2 Toric Encoded Circuits

**Definition 2.1** (Toric Encoded Circuit). Let  $C$  be a quantum circuit. We say that  $C$  is *Toric encoded* if the following holds:

1. There is a Toric code  $\mathcal{C}$  such that the qubits of  $\mathcal{C}$  can be partitioned into registers  $R_1, \dots, R_k$ , and each of the registers is encoded by  $\mathcal{C}$ .
2. The gates in  $C$  with even time coordinates are sweep decoding gates.

For a set of faults  $\mathcal{E}$  the gates in  $C[\mathcal{E}]$  can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, \dots, D_1 P_1 \Lambda_1$$

**Definition 2.2** (Base graph  $G_o$ ). Let  $C$  be a quantum circuit, at depth  $d$ , with registers  $R_1, \dots, R_k$ , all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by  $\mathcal{G}_1, \dots, \mathcal{G}_k$  the Toric complexes that induce the codes. We define the base graph of  $C$ , denoted by  $G_o = (V_o, E_o)$  as follows:

1. The vertices  $V_o$  are tuples, where the first coordinates correspond to a vertex in the union over the vertices in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges  $E_o$  are derived from the original edges in  $\mathcal{G}_1, \dots, \mathcal{G}_k$ , with the addition that any two vertices in preceding times  $t$  and  $t+1$  that support qubits (faces) such that  $C$  acts non-trivially on those qubits at time  $t$  are also connected by an edge. Namely:

$$E_o := \left\{ \{(v, t), (u, t')\} : \begin{array}{l} \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \end{array} \right\}$$

**[COMMENT]** Here we should add that a vertex  $v$  belongs to location  $l$  if there is a qubit  $f$  that  $v$  belongs to its associated face.

**Definition 2.3** (Time Graph of  $C(\mathcal{E})$ ). For a quantum circuit  $C$  and set of faults  $\mathcal{E}$ , let us denote the base graph of  $C[\mathcal{E}]$  by  $G_o$ . For each time  $t$ , denote the deviation induced by  $\mathcal{E}$  at time  $t$  by  $\mathcal{D}_t$ , so  $\partial\mathcal{D}_t$  is the syndrome that the qubits in the deviation admit. We define the history graph  $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$  as follows:

1. The vertices  $V_{\mathcal{H}}$  are taken to be the vertices of  $G_o$ , namely tuples  $(v, t) \in V_o$ , that satisfy that  $v$  belongs to a face associated with a check, which, at time  $t$ , belongs to the  $\partial\mathcal{D}_t$ . Namely:

$$\{v_t : v_t = (v, t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial\mathcal{D}_t\}$$

2. The edges  $E_{\mathcal{H}}$  is partitioned into two types  $A_{\mathcal{H}}$  and  $F_{\mathcal{H}}$ , where  $A_{\mathcal{H}}$  are the edges in  $E_o$  restricted to  $V_{\mathcal{H}}$  and  $F_{\mathcal{H}}$  are defined to be the . Namely:

$$\begin{aligned} A_{\mathcal{H}} &= \{e = \{u, v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1\} \\ F_{\mathcal{H}} &= \{e = \{u, v\} \in E_o : \text{exists } w \in V_{\mathcal{H}} \text{ such } \{v, w\}, \{u, w\} \in A_{\mathcal{H}}\} \end{aligned}$$

By definition of the construction,  $G_{\mathcal{H}}$  is a time-graph.

**Definition 2.4** (**Size()**-Function). Let  $S$  be a subset of vertices, we define the *size* of  $S$  to be:

$$\mathbf{Size}(S) := \sum_i^{d+1} \max_{v \in S} L_i(v)$$

**Claim 2.5.** Let  $\mathbf{S} = \{S_1, S_2, \dots, S_k\}$  where  $S_i \subset \mathbb{R}^d$ , introduce an edge between  $S_i$  and  $S_j$  iff  $S_i \cap S_j \neq \emptyset$ . Assume that  $\mathbf{S}$  is connected. Then  $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$ .

*Proof.* It suffices to prove that if  $R_1 \cap R_2 \neq \emptyset$ , then  $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$ . Let  $a_{i,j} = \max_{v \in R_j} L_i(v)$ . Then for any  $w \in R_1 \cap R_2$  we have:

$$\begin{aligned} \mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\ &\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\ &= \mathbf{Size}(R_1) + \mathbf{Size}(R_2) \end{aligned}$$

□

**Definition 2.6** (Size Invariant Action). [\[COMMENT\]](#) Here we classify both transversal gates, and rotation on the Toric. An action  $g$  is be said size invariant action if, for any assignment  $z$ , it holds that  $\mathbf{Size}(gz) = \mathbf{Size}(z)$ .

**Claim 2.7.** The sweep rule is Size  $\xi$ -shrinking.

*Proof.*

□

[\[COMMENT\]](#) Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time =  $3t$ , in that we apply an invariant action so.